ELECSECURE

Canadian Patent Application Package

CIPO DRAFT - VERSION 2.1

SMART ELECTRICAL SAFETY AND ENERGY EFFICIENCY MANAGEMENT SYSTEM WITH IoT-ENABLED ARC FAULT PROTECTION AND INTEGRATED CLOUD ANALYTICS

Applicant / Inventor	Azfar Mushtaq
Jurisdiction	Canada — CIPO (ISED)
Document version	Draft 2.0 — September 2026
Patent claims	20 (device, system, method)
Technical figures	11 drawings
IPC classification	H02H 3/00, H02H 1/00, H04Q 9/00

Arc fault detection & extinguishing	**Dual thermal-magnetic tripping**
IoT cloud analytics + ML	**2-module compact DIN form factor**
Mobile remote control	**Smart-home integration**

Important: Technical draft for review by a registered Canadian patent agent. Not legal advice. Not filed with CIPO.

Table of Contents

Executive Summary

ElecSecure is an integrated smart electrical safety and energy efficiency management system designed for residential and commercial installations in Canada. The invention combines a compact IoT-enabled circuit protection device with arc fault detection, active arc extinguishing, dual tripping mechanisms, and cloud-connected analytics delivered through a mobile application.

Safety Innovation	Energy Intelligence	Connected Control
Arc fault detection Arc chamber extinguishing Continuous self-test 6/10 kA breaking capacity	Real-time consumption monitoring ML efficiency recommendations Power quality analytics Historical trend reports	Wi-Fi / BLE connectivity Remote trip & reset Push alerts Smart-home API integration

FIG. 9 - INVENTION OVERVIEW

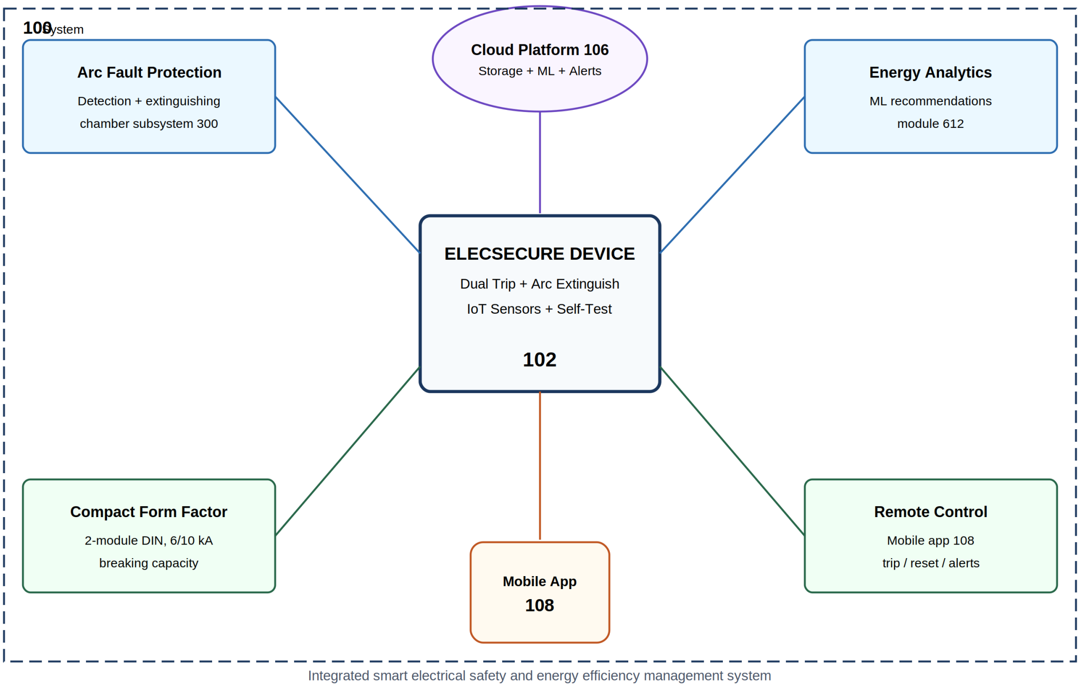


FIG. 9 — Invention overview showing integrated system (100) with core device (102), cloud platform (106), and mobile application (108).

Competitive Differentiation

The following comparison illustrates how ElecSecure differs from conventional solutions in the electrical protection and energy management market.

Feature	Conventional Breaker	Smart Meter Only	ElecSecure (Invention)
Arc fault extinguishing chamber	No	No	Yes
Dual thermal + magnetic tripping	Partial	No	Yes
Compact 2-module form factor	Varies	N/A	Yes (36 mm)
Continuous self-test + LED diagnostics	Rare	No	Yes
Real-time mobile alerts	No	Limited	Yes
Remote trip / reset control	No	No	Yes
ML energy recommendations	No	Basic	Yes
Smart-home integration	No	Limited	Yes
Bidirectional power connection	Rare	N/A	Yes

SECTION I

FILING GUIDE

Applicant: Azfar Mushtaq

Invention: Smart Electrical Safety and Energy Efficiency Management System (ElecSecure)

Jurisdiction: Canada — Canadian Intellectual Property Office (CIPO)

Document package version: Draft 1.0 — September 2026

Important legal notice

This package is a **technical draft** prepared to support a Canadian patent application. It is **not legal advice** and does **not** replace consultation with a **registered Canadian patent agent or patent lawyer**. CIPO requires compliance with the *Patent Act* and *Patent Rules* (SOR/2019-251). A professional should review claims, prior art, and formalities before filing.

Documents in this package

File	CIPO requirement	Purpose
01_PETITION.md	Petition (s. 53, Patent Rules)	Formal request for patent; inventor/applicant details
02_ABSTRACT.md	Abstract (s. 55)	Concise summary (max 150 words recommended)
03_DESCRIPTION.md	Description (s. 56)	Full technical disclosure
04_CLAIMS.md	Claims (ss. 60–62)	Legal scope of protection
05_DRAWINGS_BLUEPRINTS.md	Drawings index (s. 59)	Figure descriptions and reference numerals
drawings/*.svg	Drawings (s. 59)	11 black-and-white technical figures

Canadian filing steps

1. Prior art search (recommended before filing)

- Search CIPO patent database: https://www.ic.gc.ca/opic-cipo/cpd/eng/search/basic.html
- Search Google Patents, Espacenet, and USPTO for: arc fault circuit interrupters, IoT circuit breakers, smart energy management, dual-trip protection devices.
- Document results in a separate prior-art memorandum for your patent agent.

2. Prepare compliant application

- Each section (petition, abstract, description, claims, drawings) must **start on a new page** when filed on paper; for electronic filing via CIPO's portal, upload as separate PDF sections.
- Text: minimum 1.5 line spacing; characters ≥ 0.21 cm height.
- Drawings: black and white, A4, margins (top/left 2.5 cm, right 1.5 cm, bottom 1.0 cm); reference characters ≥ 0.32 cm.
- Language: **English** (or French — do not mix).

3. File with CIPO

Online (recommended): https://www.ic.gc.ca/opic-cipo/cpd/eng/help/patents/e-filing.html

Minimum for filing date:

- Petition
- Applicant name and address
- Description
- Claims (at least one)
- Filing fee

Submit soon after filing date:

- Abstract
- Drawings (if invention cannot be understood without them — **recommended for this invention**)
- Inventor name and address
- Statements of entitlement / inventorship

4. Fees (effective January 1, 2026 — verify on CIPO website)

Fee	Standard entity	Small entity (if eligible)
Application (filing)	$595.06 CAD	$241.24 CAD
Request for examination	$1,190.13 CAD (basic)	$482.48 CAD
Excess claims (>20)	$117.94 CAD each	$58.97 CAD each
Final fee (on allowance)	$446.03 CAD (basic)	$181.20 CAD

Small entity status: generally Canadian businesses with ≤100 employees and qualifying revenue; confirm eligibility with your agent.

5. Request examination

- Examination must be **requested within 4 years** of the filing date (or PCT national phase entry).
- Without a request for examination, the application will be deemed abandoned.

6. Prosecution timeline (typical)

- First Office Action: often 2–4 years after examination request.
- Respond to objections within **4 months** (extendable).
- Patent term: **20 years** from Canadian filing date (subject to maintenance fees).

Inventorship and ownership

Role	Name	Notes
Inventor	Azfar Mushtaq	Complete inventive contribution statement with agent
Applicant	[To be confirmed — individual or corporation]	Assignments may be required if applicant ≠ inventor

Recommended filing strategy

1. **Canadian national application** — file directly with CIPO using this package.
2. **PCT international application** — if protection in multiple countries is planned within 12 months, file PCT claiming priority from the Canadian filing.
3. **Provisional (US only)** — Canada has no provisional application; early Canadian filing establishes priority.

Post-filing checklist

- [ ] Record official filing date and application number
- [ ] Calendar: request examination before 4-year deadline
- [ ] Calendar: pay maintenance fees (annually from 2nd anniversary)
- [ ] File trademark for "ElecSecure" separately (CIPO Trademarks)
- [ ] Register copyright for mobile app source code (automatic; optional registration with CIPO)
- [ ] Maintain invention as trade secret for unfiled embodiments until filing

Contact — CIPO

Canadian Intellectual Property Office

Innovation, Science and Economic Development Canada

https://ised-isde.canada.ca/site/canadian-intellectual-property-office/en

Patent Information Centre: 1-866-997-1936 (toll-free in Canada)

Revision log

Version	Date	Author	Changes
1.0	2026-09-01	Draft for Azfar Mushtaq	Initial Canadian patent package

SECTION II PETITION

TO THE COMMISSIONER OF PATENTS

The undersigned hereby applies for a patent for an invention entitled:

SMART ELECTRICAL SAFETY AND ENERGY EFFICIENCY MANAGEMENT SYSTEM WITH IoT-ENABLED ARC FAULT PROTECTION AND INTEGRATED CLOUD ANALYTICS

Applicant

Field	Details
Name	Azfar Mushtaq
Address	[Street Address]
City	[City]
Province	[Province]
Postal Code	[Postal Code]
Country	Canada
Telephone	[Phone]
Email	[Email]

If applicant is a corporation, substitute corporate name and registered office address.

Inventor(s)

#	Name	Residence	Citizenship
1	Azfar Mushtaq	[City, Province, Canada]	[Nationality]

Add co-inventors if applicable. Each inventor's postal address is required by s. 54, Patent Rules.

Statement of Entitlement

The applicant is entitled to apply for and be granted a patent for the invention because:

- [ ] The applicant is the inventor.
- [ ] The applicant is the legal representative of the inventor.
- [ ] The applicant is entitled to apply by virtue of an assignment from the inventor(s).
- [ ] Other: ________________________________

Statement of Inventorship

The person(s) named above as inventor(s) is/are the actual inventor(s) or the legal representative(s) of the actual inventor(s) of the subject-matter defined in the claims.

Priority claim (if applicable)

Earlier application	Filing date	Country
None claimed at this time	—	—

If filing within 12 months of an earlier application, complete priority details and attach certified copy if requested.

Agent (if represented)

Field	Details
Registered patent agent name	[To be appointed]
Agent registration number	[CIPO agent number]
Firm	[Patent agency / law firm]
Address	[Address]

Declaration

I/We declare that:

1. I am/we are the applicant(s) for the patent identified above.
2. The statements made in this petition are true and complete.
3. I/we have reviewed the description, claims, abstract, and drawings submitted with this application.
4. I/we authorize CIPO to correspond with the agent named above (if any).

Signature: ________________________________

Name: Azfar Mushtaq

Date: ________________________________

Capacity: Applicant / Authorized representative

This petition conforms to section 53 of the Patent Rules (SOR/2019-251). Submit as a separate document starting on a new page when filing with CIPO.

SECTION III ABSTRACT

A smart electrical safety and energy efficiency management system comprises an IoT-enabled protective electrical device (ElecSecure), a cloud-based data platform, and a mobile application. The protective device integrates dual tripping mechanisms—a delayed thermal trip for overload conditions and a magnetic trip for short-circuit conditions—with arc fault detection and arc extinguishing via guided arc runners and a multi-plate arc chamber. The device occupies a compact two-module form factor while providing selectable high breaking capacity of 6 kA or 10 kA. Continuous internal self-test, LED fault indication, and bidirectional power connection enhance reliability and serviceability. On-board sensors measure current, voltage, temperature, and arc signatures; a microcontroller processes fault conditions locally and transmits telemetry via Wi-Fi or Bluetooth Low Energy to a cloud platform. The platform stores historical data, generates real-time alerts, and applies machine-learning algorithms to identify energy wastage and deliver personalized efficiency recommendations. A mobile application provides remote monitoring, circuit breaker trip/reset control, data visualization, and integration with smart-home ecosystems. The integrated system improves electrical safety, reduces fire risk from arcing faults, and optimizes energy consumption for residential and commercial installations.

Word count: 149

Abstract must begin on a new page. Must not contain drawings or reference to figure numbers as sole support for features.

SECTION IV DESCRIPTION

TITLE OF THE INVENTION

SMART ELECTRICAL SAFETY AND ENERGY EFFICIENCY MANAGEMENT SYSTEM WITH IoT-ENABLED ARC FAULT PROTECTION AND INTEGRATED CLOUD ANALYTICS

TECHNICAL FIELD

The present invention relates to electrical protection and energy management systems. More particularly, the invention relates to an integrated system combining an IoT-enabled circuit protection device with arc fault detection and extinguishing, dual tripping mechanisms, continuous self-testing, and cloud-connected analytics for real-time electrical safety monitoring and energy efficiency optimization in residential and commercial installations.

BACKGROUND OF THE INVENTION

Electrical distribution systems in buildings require protection against overload, short-circuit, and arcing fault conditions. Conventional circuit breakers and residual-current devices provide basic overcurrent and ground-fault protection but often lack integrated diagnostics, remote monitoring, and energy analytics capabilities.

Arcing faults—unintentional electrical discharges across an air gap—are a leading cause of residential and commercial electrical fires. Conventional thermal-magnetic breakers may not detect low-level series arcing faults before significant damage occurs. Dedicated arc-fault circuit interrupters (AFCIs) address some arc conditions but are typically deployed as separate devices without integrated energy management or cloud connectivity.

The proliferation of smart home technology and rising energy costs have increased demand for systems that combine electrical safety with real-time energy monitoring, remote control, and data-driven efficiency recommendations. Existing market offerings generally address only subsets of these requirements: standalone smart meters provide consumption data without integrated protection; conventional breakers provide protection without IoT connectivity; and energy management platforms lack embedded arc fault extinguishing hardware in a compact form factor.

There remains a need for a unified system that:

1. Integrates dual thermal-magnetic tripping with active arc fault detection and arc extinguishing in a compact protective device;
2. Provides continuous internal self-testing and localized LED fault indication;
3. Connects securely to a cloud platform for real-time telemetry, alerts, and historical analytics;
4. Delivers machine-learning-based energy efficiency recommendations through a mobile application;
5. Enables remote trip/reset control and smart-home ecosystem integration.

The present invention addresses these needs.

SUMMARY OF THE INVENTION

In a first aspect, the invention provides an IoT-enabled electrical protection device comprising:

- a housing occupying a two-module DIN-rail form factor;
- a delayed thermal tripping mechanism configured to interrupt current flow upon sustained overload;
- a magnetic tripping mechanism configured to interrupt current flow upon short-circuit;
- an arc fault detection subsystem comprising at least one arc sensor and signal processing circuitry;
- an arc extinguishing subsystem comprising arc runners, pre-chamber plates, and an arc chamber configured to divide an electric arc into progressively smaller arc segments until extinguished;
- a microcontroller configured to execute local fault analysis, continuous internal self-test routines, and communication with a connectivity module;
- a connectivity module supporting at least one of Wi-Fi and Bluetooth Low Energy;
- a sensor array comprising current sensors, voltage sensors, and temperature sensors;
- an LED indication subsystem for localized fault identification; and
- a bidirectional power connection interface permitting supply connection from either a top or bottom terminal orientation.

In a second aspect, the invention provides a smart electrical safety and energy efficiency management system comprising the IoT-enabled electrical protection device, a cloud-based data platform, and a mobile application, wherein the cloud platform receives telemetry from one or more devices, stores historical electrical parameter data, generates real-time alerts upon detection of abnormal conditions, and applies machine-learning algorithms to produce personalized energy efficiency recommendations transmitted to the mobile application.

In a third aspect, the invention provides a method of managing electrical safety and energy efficiency comprising: acquiring electrical parameter data from sensors in an IoT-enabled protection device; performing local arc fault detection and tripping when arc signatures exceed predetermined thresholds; transmitting processed telemetry to a cloud platform; analyzing historical and real-time data using machine-learning models; generating alerts and efficiency recommendations; and presenting data and control functions to a user via a mobile application including remote trip and reset commands.

Further aspects and embodiments are defined in the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the invention will now be described by way of example with reference to the accompanying drawings, in which:

Figure 1 is a system architecture diagram showing the ElecSecure protection device, connectivity layer, cloud platform, mobile application, and smart-home integrations.

Figure 2 is a front elevational view of the ElecSecure device showing module dimensions, terminals, and LED indicators.

Figure 3 is a sectional side view of the arc extinguishing subsystem showing arc runners (302), pre-chamber plates (304), and arc chamber (306).

Figure 4 is a functional block diagram of internal electronics including microcontroller (402), tripping mechanisms (404, 406), sensors (408), connectivity module (410), and power supply (412).

Figure 5 is a flowchart of the local fault detection and arc extinguishing process.

Figure 6 is a flowchart of cloud data processing, machine-learning analysis, and mobile application interaction.

Figure 7 is a schematic diagram of dual tripping mechanism operation under overload and short-circuit conditions.

Figure 8 is a user interface wireframe of the mobile application showing real-time monitoring, alerts, and remote control panels.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

System Overview (Figure 1)

Reference numeral **100** designates the smart electrical safety and energy efficiency management system. System **100** comprises an ElecSecure protection device **102**, a local network **104**, a cloud platform **106**, a mobile application **108**, and optional smart-home devices **110**.

The protection device **102** is installed in an electrical distribution panel and monitors one or more circuits. Device **102** acquires electrical parameters, performs local protection functions including arc fault detection and extinguishing, and transmits telemetry via connectivity module **112** through local network **104** to cloud platform **106**.

Cloud platform **106** aggregates data from multiple devices **102**, performs storage, analytics, and machine-learning processing, and pushes alerts and recommendations to mobile application **108**. Mobile application **108** provides user interfaces for monitoring, reporting, remote control, and integration with smart-home ecosystems **110**.

ElecSecure Protection Device — Mechanical Design (Figure 2)

The protection device **102** is housed in an enclosure **202** conforming to a two-module DIN-rail form factor, providing a width of approximately 36 mm (two standard 18 mm modules). Despite the compact footprint, the device supports selectable breaking capacities of **6 kA** or **10 kA** in accordance with applicable product standards (e.g., IEC 60898, IEC 61009, or CSA C22.2 No. 5 as adapted for Canadian installations).

Input terminals **204** and output terminals **206** are arranged to permit bidirectional power connection—electrical supply may be connected to either the top or bottom terminal pair, improving installation flexibility in constrained panel layouts.

LED indicators **208** provide localized fault identification. Each LED corresponds to a fault category (e.g., overload, short-circuit, arc fault, ground fault, communication loss, self-test failure), enabling field technicians to diagnose conditions without requiring immediate access to the mobile application.

A manual test/reset interface **210** permits user-initiated self-test cycles and manual reset following a trip event.

Arc Extinguishing Subsystem (Figure 3)

The arc extinguishing subsystem **300** is a distinguishing feature of device **102**. When a fault condition produces an electric arc, arc runners **302** guide the arc from the contact separation point into a pre-chamber defined by pre-chamber plates **304**. The arc is directed into arc chamber **306**, which is configured with insulating partitions and cooling surfaces that subdivide the main arc into a series of smaller arc segments.

Each subdivision increases arc voltage drop and reduces arc current until the arc is extinguished. This active arc management reduces contact erosion, limits thermal damage to adjacent components, and decreases the probability of arc-induced ignition of surrounding materials.

Arc sensors **308** (optical, acoustic, or current-signature-based) detect arc initiation. Upon detection, the microcontroller **402** (Figure 4) commands trip actuation within a predetermined response time (typically less than 100 ms for series arc conditions, subject to applicable standard requirements).

Internal Electronics (Figure 4)

The electronic architecture of device **102** centers on microcontroller **402**, which may be implemented as an ARM Cortex-M class processor or equivalent embedded controller with sufficient processing capacity for real-time sensor sampling, arc signature analysis, and communication protocol handling.

Delayed thermal tripping mechanism 404 comprises a bimetallic element or equivalent thermal actuator calibrated to trip after a sustained overcurrent condition exceeding the rated current by a time-dependent margin (inverse-time characteristic).

Magnetic tripping mechanism 406 comprises a solenoid coil that actuates instantaneously when current exceeds a magnetic trip threshold indicative of a short-circuit condition.

Sensor array **408** includes:

• Current sensors** (e.g., shunt resistors, current transformers, or Hall-effect sensors) for per-circuit amperage measurement;

• Voltage sensors** for line voltage and power quality monitoring;

• Temperature sensors** for thermal monitoring of internal components and ambient panel temperature;

• Arc sensors** as described above.

Connectivity module **410** supports Wi-Fi (IEEE 802.11 b/g/n) for LAN communication and Bluetooth Low Energy (BLE) for provisioning, local diagnostics, and fallback communication. Firmware in non-volatile memory **414** implements encrypted TLS communication to cloud platform **106**.

Power supply **412** derives low-voltage DC power from the monitored circuit or an auxiliary supply, with voltage regulators maintaining stable operating voltage for electronics independent of line voltage fluctuations within rated range.

Continuous Internal Self-Test

Device **102** executes continuous internal self-test routines without requiring user intervention. Self-test procedures include:

1. Verification of sensor calibration within acceptable drift limits;
2. Functional test of trip actuators using simulated drive signals (non-destructive);
3. Communication module heartbeat verification;
4. Memory integrity checks (CRC on firmware and configuration);
5. Arc sensor sensitivity verification using injected test signals.

Failure of any self-test sub-routine is logged locally and reported to cloud platform **106**, and a corresponding LED indicator **208** is illuminated.

Local Fault Detection Process (Figure 5)

Process **500** begins at step **502** with continuous acquisition of sensor data. At step **504**, microcontroller **402** evaluates current magnitude against thermal and magnetic trip thresholds.

If a short-circuit threshold is exceeded (step **506**—yes), magnetic trip mechanism **406** is actuated at step **508**. If an overload condition is detected (step **510**—yes), thermal trip mechanism **404** is actuated at step **512**.

Parallel to overcurrent evaluation, arc signature analysis is performed at step **514**. Arc signatures may be characterized by high-frequency current perturbations, optical emissions, or acoustic emissions. If an arc fault is detected (step **516**—yes), the arc extinguishing subsystem **300** is engaged at step **518** concurrently with trip actuation.

At step **520**, fault data including timestamp, fault type, magnitude, and waveform excerpts are packaged for transmission. At step **522**, an alert is pushed to cloud platform **106** and mobile application **108**.

Cloud Platform and Machine Learning (Figure 6)

Process **600** describes server-side operations. At step **602**, cloud platform **106** receives encrypted telemetry from device **102**. Data is validated and stored in time-series database **604**.

At step **606**, real-time rule engine evaluates incoming data against user-configurable thresholds (voltage sag, swell, imbalance, demand peaks). Alerts generated at step **608** are delivered via push notification to mobile application **108**.

At step **610**, machine-learning module **612** analyzes historical consumption patterns. Models may include:

- Clustering of daily load profiles to identify anomalous consumption;
- Regression models predicting demand peaks;
- Classification models distinguishing normal inrush current from fault conditions to reduce nuisance tripping;
- Recommendation engine suggesting load scheduling, standby power reduction, and equipment maintenance.

Recommendations are transmitted to mobile application **108** at step **614. User feedback on recommendations (accepted, dismissed, implemented) is collected at step** 616** to improve model accuracy through supervised learning.

Dual Tripping Mechanism (Figure 7)

Figure 7 illustrates operation of dual tripping mechanisms under two fault regimes.

Under **overload condition 702**, current **704** exceeds rated current **706** for duration **708** sufficient to heat bimetallic element **710**, causing mechanical displacement **712** that releases latch **714** and separates contacts **716**.

Under **short-circuit condition 720**, current **722** rises rapidly above magnetic trip threshold **724**, energizing solenoid **726** to actuate instantaneous trip **728** of contacts **716**.

Both mechanisms operate independently, ensuring appropriate response to fault type without compromising selectivity.

Mobile Application (Figure 8)

Mobile application **108** provides:

- Dashboard 802**: real-time voltage, current, power, power factor, and energy consumption;
- Circuit panel 804**: per-circuit status with color-coded health indicators;

• Alert center 806**: chronological fault and warning notifications with severity classification;

• Remote control 808**: authorized trip and reset commands with two-factor authentication;

• Analytics 810**: graphs, trend reports, and exportable CSV/PDF reports;

• Recommendations 812**: machine-learning-generated efficiency suggestions;

• Smart-home integration 814**: APIs for coordination with thermostats, lighting, and home automation hubs.

Smart Home Integration

Application **108** exposes a RESTful API and/or MQTT interface for integration with smart-home platforms. Users may define automation rules such as:

• Shedding non-critical loads upon detection of impending demand threshold;

• Sending notifications to building management systems upon arc fault events;

• Coordinating backup generator transfer upon utility voltage loss detection.

Earthing and Lightning Protection (Optional Embodiment)

In an extended embodiment, system **100** includes earthing and lightning protection components **120** compliant with IEC 62305 and Canadian Electrical Code (CEC) Part I requirements. These components integrate with cloud platform **106** to report ground resistance monitoring values and surge protection device status.

Security

All communication between device **102**, cloud platform **106**, and mobile application **108** employs:

• TLS 1.2 or higher encryption in transit;

• Mutual authentication using device certificates;

• Role-based access control for remote trip/reset functions;

• Data compression for bandwidth optimization without compromising encrypted payload integrity.

Canadian Regulatory Considerations

For deployment in Canada, device **102** is designed for compliance with:

• Canadian Electrical Code (CEC), Part I (CSA C22.1);

• CSA C22.2 No. 5 — Molded-case circuit breakers;

• Applicable AFCI requirements per CEC Rule 26-720 et seq. for dwelling unit branch circuits;

• Innovation, Science and Economic Development Canada (ISED) radio equipment certification for Wi-Fi/BLE modules.

EXAMPLES

Example 1 — Residential Panel Installation

A single ElecSecure device **102** is installed on a 15 A branch circuit serving kitchen receptacles. During operation, a deteriorating connection on a countertop appliance produces series arcing. Arc sensor **308** detects characteristic high-frequency current modulations. Microcontroller **402** actuates trip within 80 ms. Arc extinguishing subsystem **300** limits arc duration. Mobile application **108** delivers push notification: "Arc fault detected — Circuit 7 tripped." Cloud analytics identify the circuit as a recurring fault location, recommending professional inspection.

Example 2 — Commercial Energy Optimization

A facility deploys twelve devices **102** across distribution panels. Cloud platform **106** aggregates consumption data. Machine-learning module **612** identifies that HVAC loads operate concurrently with peak tariff periods. Application **108** recommends rescheduling two circuits to off-peak hours, projecting 8% monthly cost reduction.

INDUSTRIAL APPLICABILITY

The invention is applicable to residential, commercial, and light-industrial electrical installations requiring enhanced arc fault protection, real-time monitoring, and energy efficiency management. The compact form factor enables retrofit installation in existing panels without significant spatial modification.

End of Description. This document must begin on a new page when filed. Reference numerals in this description correspond to figures in the Drawings section.

SECTION V CLAIMS

Claims

What is claimed is:

1. An IoT-enabled electrical protection device comprising:

- a housing configured for mounting on a DIN rail in a two-module form factor;
- a delayed thermal tripping mechanism configured to interrupt electrical current upon detection of a sustained overload condition;
- a magnetic tripping mechanism configured to interrupt electrical current upon detection of a short-circuit condition;
- an arc fault detection subsystem comprising at least one arc sensor and processing circuitry configured to identify an arcing fault based on at least one of current signature, optical emission, or acoustic emission;
- an arc extinguishing subsystem comprising arc runners, pre-chamber plates, and an arc chamber configured to receive an electric arc from the arc runners and to subdivide the electric arc into a plurality of smaller arc segments until the arc is extinguished;
- a microcontroller operatively coupled to the arc fault detection subsystem, the delayed thermal tripping mechanism, and the magnetic tripping mechanism, the microcontroller configured to execute local fault analysis and continuous internal self-test routines;
- a sensor array comprising at least a current sensor and a voltage sensor;
- a connectivity module operatively coupled to the microcontroller and configured to communicate over at least one wireless protocol selected from Wi-Fi and Bluetooth Low Energy; and
- an LED indication subsystem configured to indicate a fault category upon occurrence of a fault condition.

2. The device of claim 1, wherein the housing provides a selectable breaking capacity of at least one of 6 kA or 10 kA.

3. The device of claim 1 or 2, further comprising a bidirectional power connection interface permitting electrical supply connection to either a top terminal pair or a bottom terminal pair.

4. The device of any one of claims 1 to 3, wherein the continuous internal self-test routines comprise at least one of: sensor calibration verification, trip actuator functional verification, communication module heartbeat verification, memory integrity verification, and arc sensor sensitivity verification.

5. The device of any one of claims 1 to 4, wherein the arc fault detection subsystem is configured to command trip actuation within a predetermined response time upon detection of an arcing fault.

6. The device of any one of claims 1 to 5, further comprising a temperature sensor in the sensor array for monitoring at least one of internal component temperature or ambient panel temperature.

7. A smart electrical safety and energy efficiency management system comprising:

- the IoT-enabled electrical protection device of any one of claims 1 to 6;
- a cloud-based data platform configured to receive telemetry from the IoT-enabled electrical protection device, store historical electrical parameter data, and generate alerts upon detection of

abnormal electrical conditions; and

- a mobile application configured to communicate with the cloud-based data platform to present real-time electrical parameter data, fault alerts, and remote control commands including at least trip and reset commands for the IoT-enabled electrical protection device.

8. The system of claim 7, wherein the cloud-based data platform comprises a machine-learning module configured to analyze historical and real-time electrical parameter data and to generate personalized energy efficiency recommendations transmitted to the mobile application.

9. The system of claim 7 or 8, wherein the mobile application is configured to integrate with a smart-home ecosystem via at least one of a RESTful API and an MQTT interface to coordinate electrical system functions with at least one external smart-home device.

10. The system of any one of claims 7 to 9, wherein communication between the IoT-enabled electrical protection device, the cloud-based data platform, and the mobile application is encrypted using TLS 1.2 or higher.

11. The system of any one of claims 7 to 10, wherein the mobile application presents data visualization including at least one of graphs, trend reports, and exportable consumption reports derived from telemetry stored on the cloud-based data platform.

12. The system of any one of claims 7 to 11, further comprising earthing and lightning protection components operatively coupled to the cloud-based data platform and configured to report at least one of ground resistance monitoring values and surge protection device status.

13. A method of managing electrical safety and energy efficiency, the method comprising:

- acquiring electrical parameter data from a sensor array of an IoT-enabled electrical protection device installed in an electrical distribution panel;
- performing local fault analysis on the acquired electrical parameter data using a microcontroller of the IoT-enabled electrical protection device;
- detecting an arcing fault based on at least one arc signature derived from the acquired electrical parameter data;
- upon detection of the arcing fault, actuating an arc extinguishing subsystem comprising arc runners, pre-chamber plates, and an arc chamber to extinguish an electric arc and interrupt current flow;
- transmitting processed telemetry from the IoT-enabled electrical protection device to a cloud-based data platform via a wireless connectivity module;
- analyzing at least one of historical and real-time telemetry at the cloud-based data platform using a machine-learning module to generate energy efficiency recommendations;
- generating a fault alert upon detection of an abnormal electrical condition; and
- presenting the energy efficiency recommendations, fault alerts, and remote control functions to a user via a mobile application communicatively coupled to the cloud-based data platform.

14. The method of claim 13, further comprising executing continuous internal self-test routines on the IoT-enabled electrical protection device and reporting self-test results to the cloud-based data platform.

15. The method of claim 13 or 14, further comprising:

- detecting a sustained overload condition and actuating a delayed thermal tripping mechanism; and
- detecting a short-circuit condition and actuating a magnetic tripping mechanism independently of the delayed thermal tripping mechanism.

16. The method of any one of claims 13 to 15, further comprising illuminating an LED indicator on the IoT-enabled electrical protection device to indicate a fault category corresponding to the detected abnormal electrical condition.

17. The method of any one of claims 13 to 16, further comprising receiving user feedback on the energy efficiency recommendations at the cloud-based data platform and updating the machine-learning module based on the user feedback.

18. The method of any one of claims 13 to 17, further comprising integrating the mobile application with a smart-home ecosystem to execute an automation rule triggered by at least one of the fault alert and the energy efficiency recommendations.

19. The device of any one of claims 1 to 6, wherein the at least one wireless protocol comprises Wi-Fi for local area network communication and Bluetooth Low Energy for device provisioning.

20. The system of any one of claims 7 to 12, wherein the cloud-based data platform applies a real-time rule engine to evaluate incoming telemetry against user-configurable thresholds including at least one of voltage sag, voltage swell, phase imbalance, and demand peak conditions.

SECTION VI **TECHNICAL DRAWINGS**

Technical Blueprints & Reference Numerals

Numeral	Element Description
100	Smart electrical safety and energy efficiency management system
102	ElecSecure IoT-enabled protection device
104	Local network (Wi-Fi / LAN)
106	Cloud-based data platform
108	Mobile application
110	Smart-home devices
202	Device enclosure (two-module DIN)
204/206	Bidirectional input/output terminals
208	LED fault indicators
300	Arc extinguishing subsystem
302	Arc runners
304	Pre-chamber plates
306	Arc chamber
308	Arc sensors
402	Microcontroller
404	Delayed thermal tripping mechanism
406	Magnetic tripping mechanism
408	Sensor array (current, voltage, temperature)
410	Connectivity module (Wi-Fi / BLE)
412	Power supply
414	Non-volatile memory (firmware, logs)
500	Local fault detection process
600	Cloud processing process
612	Machine-learning module

Numeral	Element Description
802–814	Mobile application UI panels

FIGURE 1

System Architecture

FIG. 1

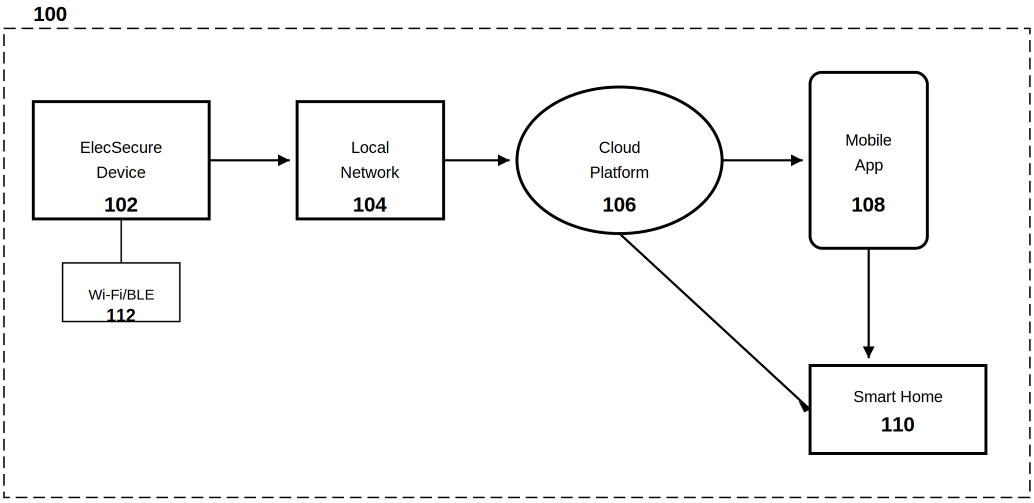


FIG. 1 - End-to-end data flow among protection device (102), local network (104), cloud platform (106), mobile app (108), and smart-home devices (110).

FIGURE 2

Device Front Elevation

FIG. 2

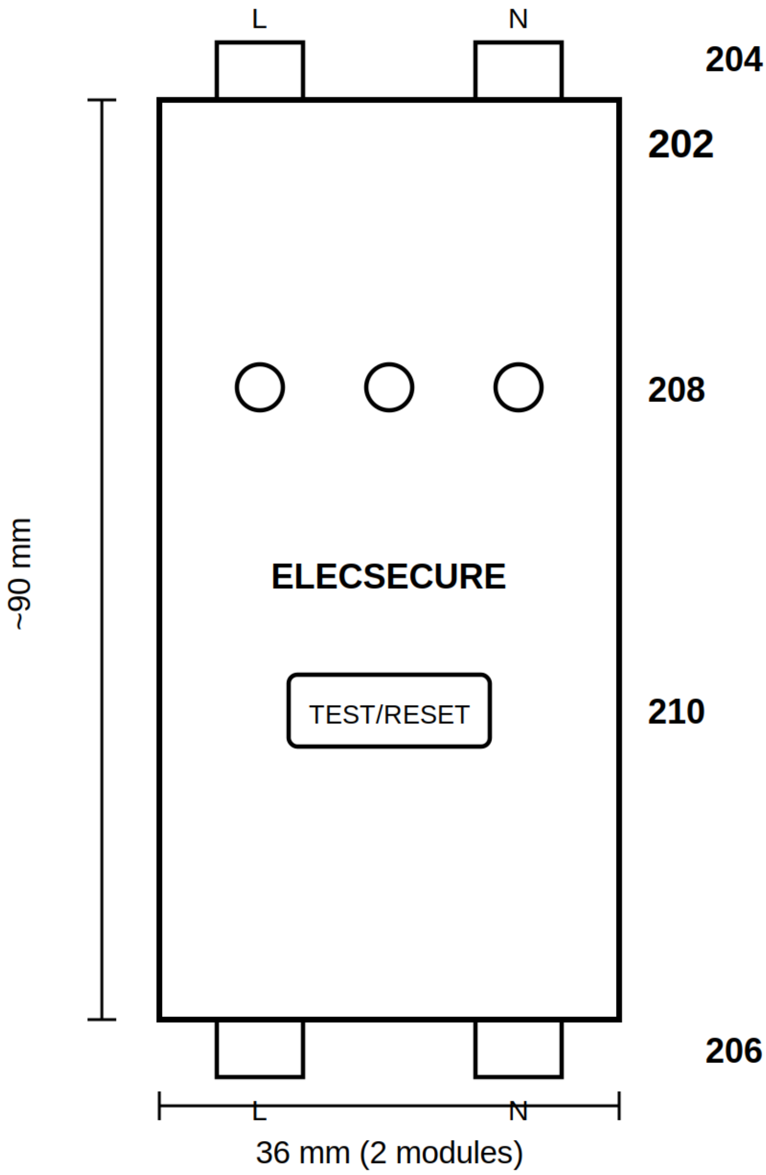


FIG. 2 - Two-module DIN form factor (202) with bidirectional terminals (204, 206), LED fault indicators (208), and test/reset interface (210). Width: 36 mm.

FIGURE 3

Arc Extinguishing Subsystem

FIG. 3

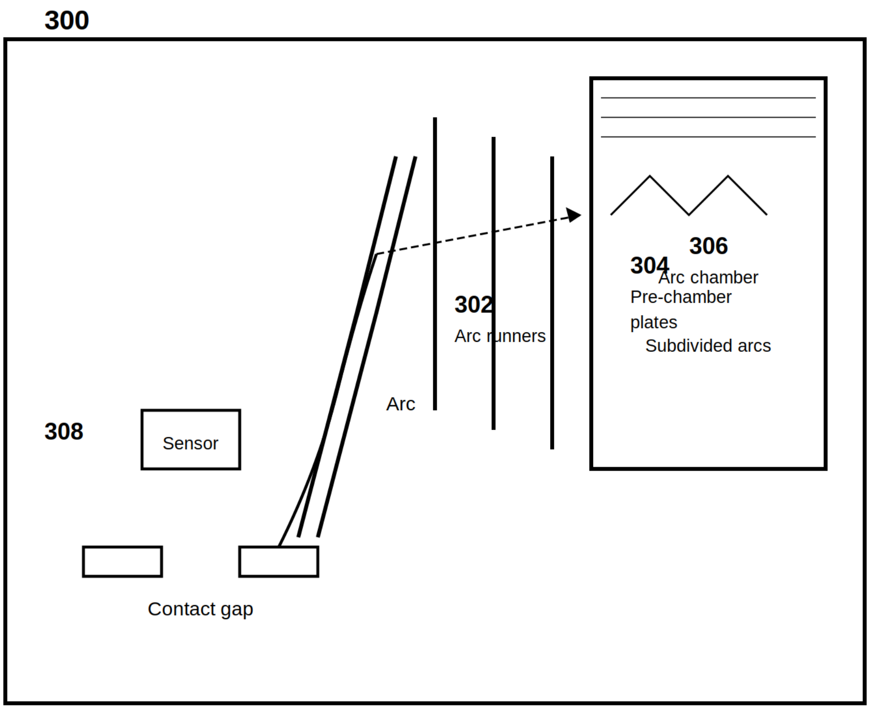


FIG. 3 - Sectional view: arc runners (302), pre-chamber plates (304), arc chamber (306), arc sensor (308).

FIGURE 4

Internal Electronics Block Diagram

FIG. 4

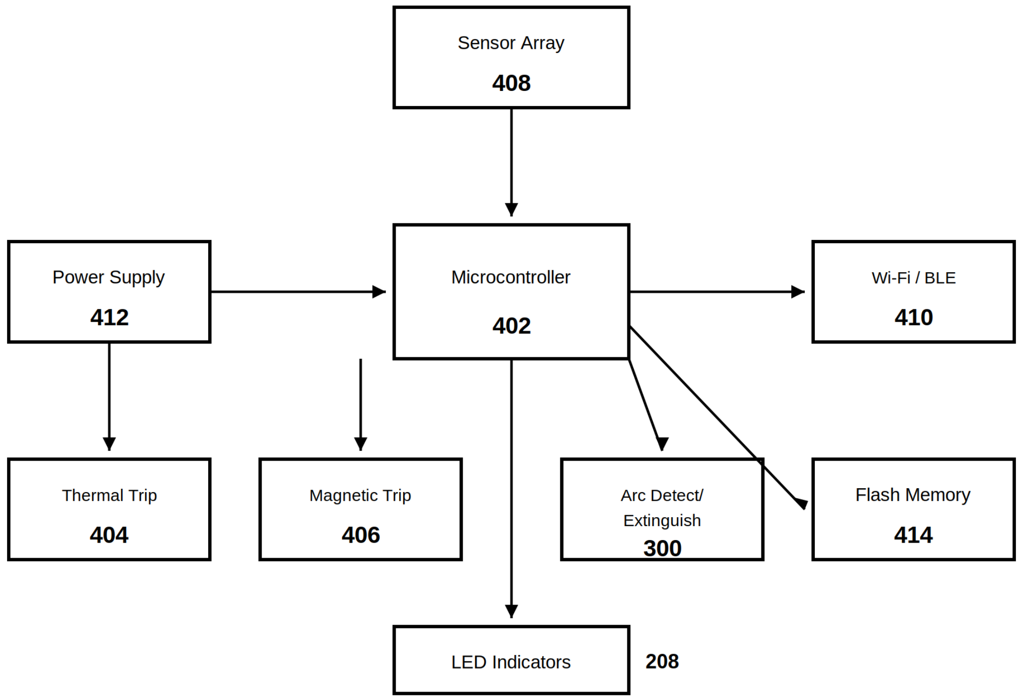
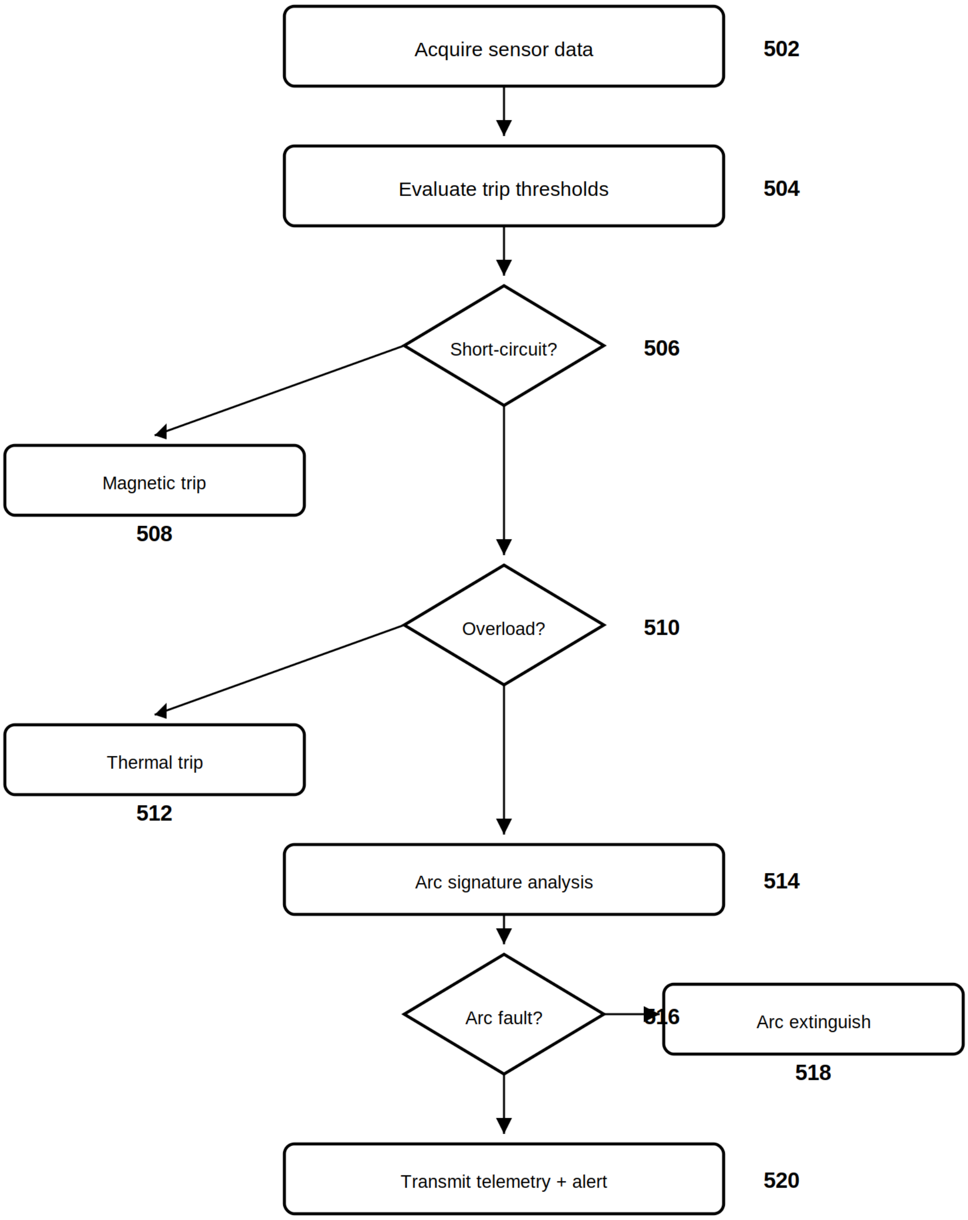


FIG. 4 - Microcontroller (402), sensors (408), trip mechanisms (404, 406), connectivity (410), memory (414), arc subsystem (300).

FIGURE 5

Local Fault Detection Flowchart

FIG. 5



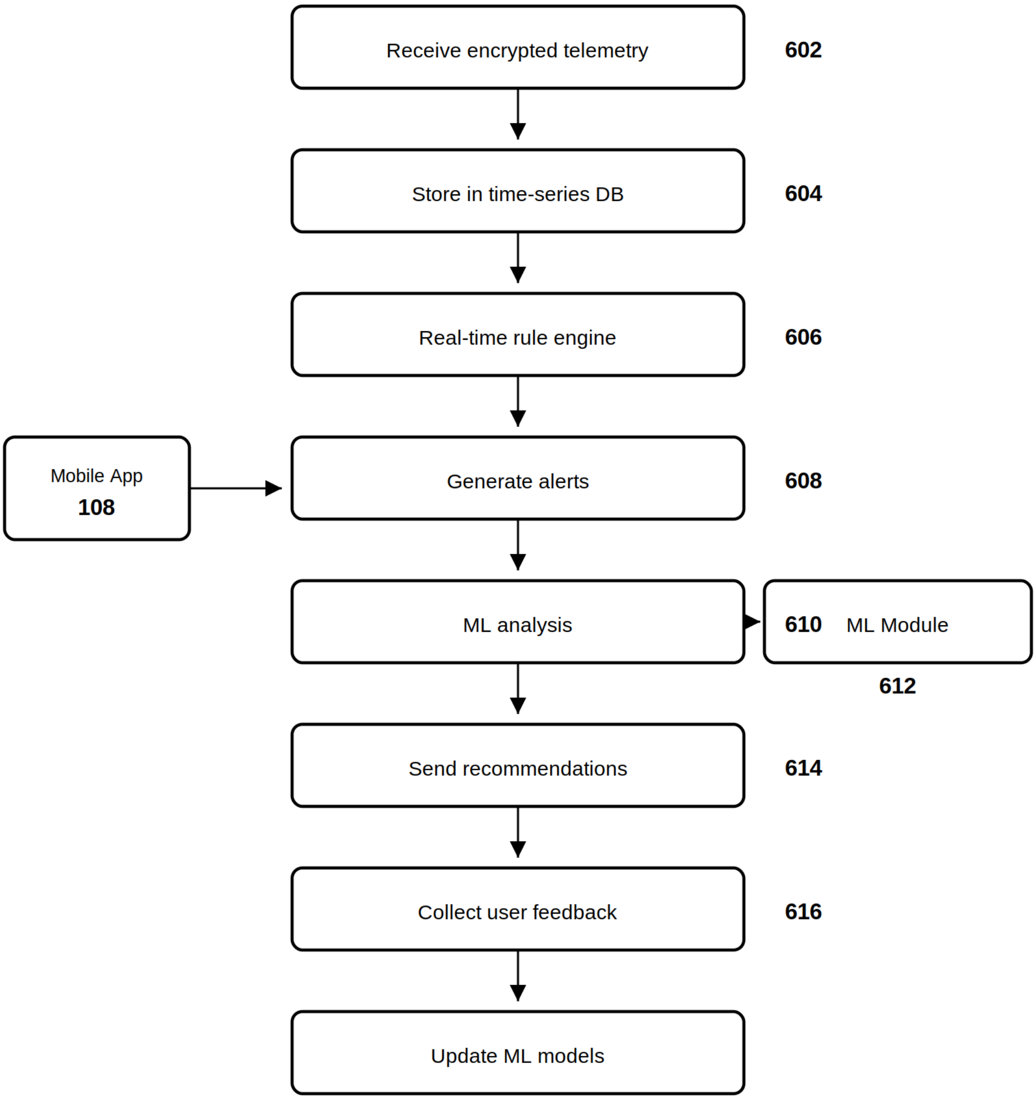
500

FIG. 5 - Process (500): acquisition, threshold evaluation, tripping, arc analysis, extinguishing, alert transmission.

FIGURE 6

Cloud Processing Flowchart

FIG. 6



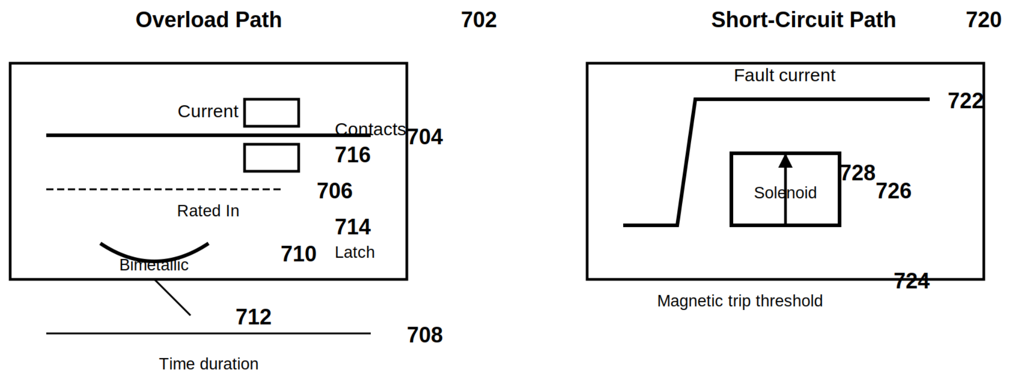
600

FIG. 6 - Process (600): ingestion, storage, rule engine, ML module (612), recommendations, feedback loop.

FIGURE 7

Dual Tripping Mechanism

FIG. 7



Thermal = inverse-time response | Magnetic = instantaneous response

Mechanisms 404 (thermal) and 406 (magnetic) operate independently

FIG. 7 - Overload path (702) via bimetallic element (710); short-circuit path (720) via solenoid (726).

FIGURE 8

Mobile Application Interface

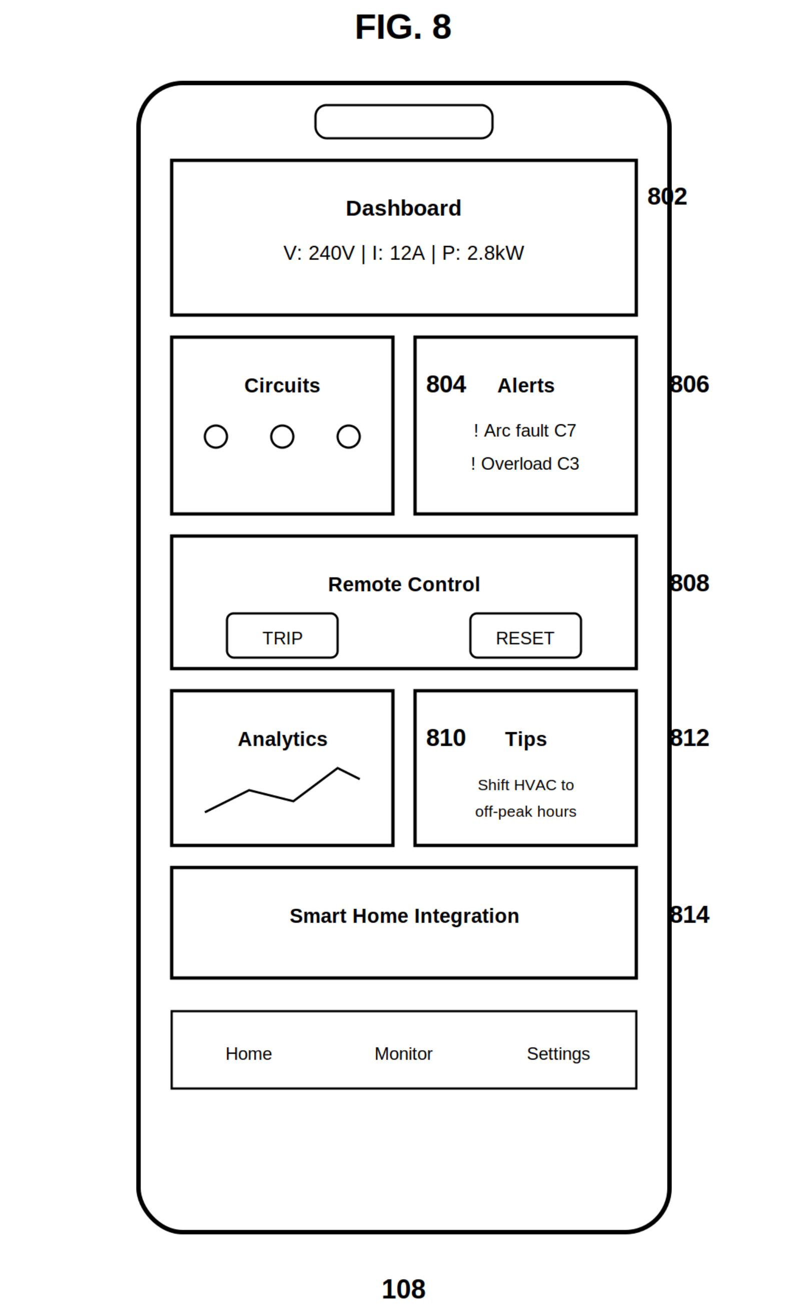


FIG. 8 - Dashboard (802), circuits (804), alerts (806), remote control (808), analytics (810), tips (812), smart-home (814).

FIGURE 9

Invention Overview

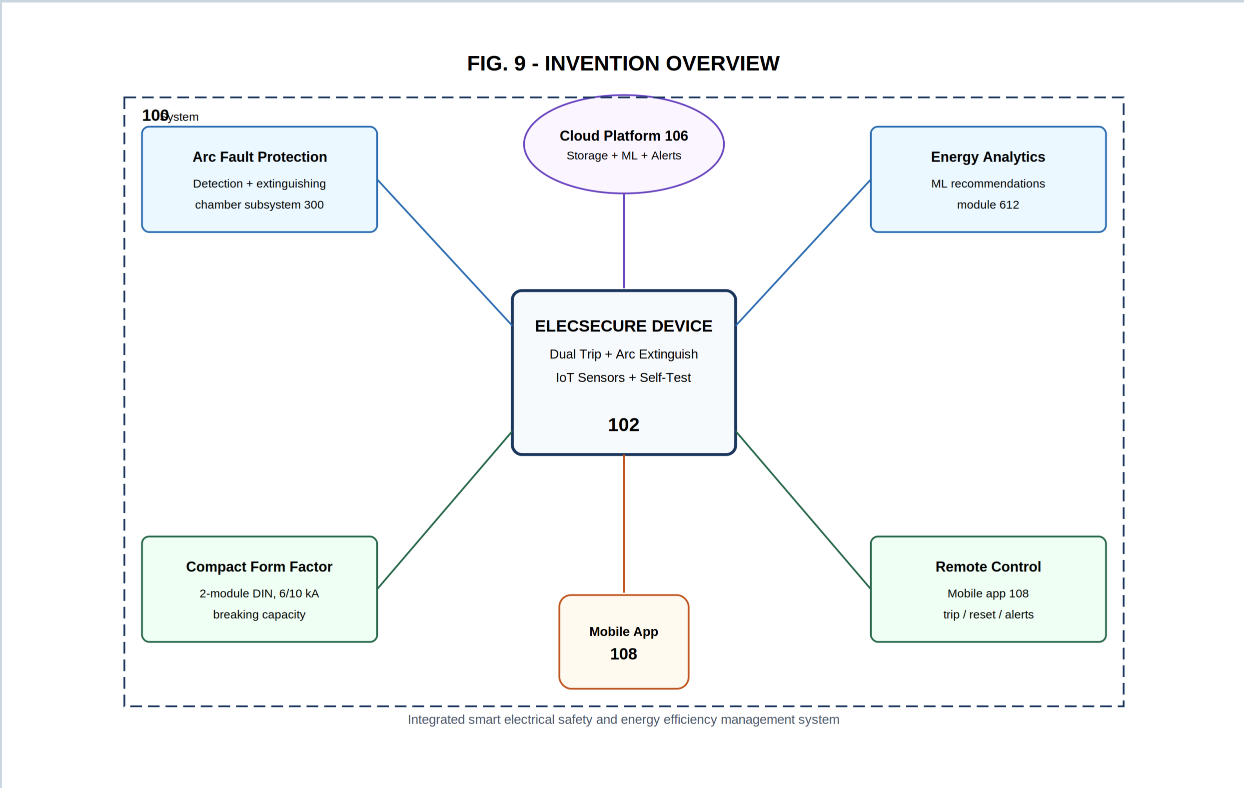


FIG. 9 - Integrated system (100) showing core inventive features and component relationships.

FIGURE 10

Panel Installation Blueprint

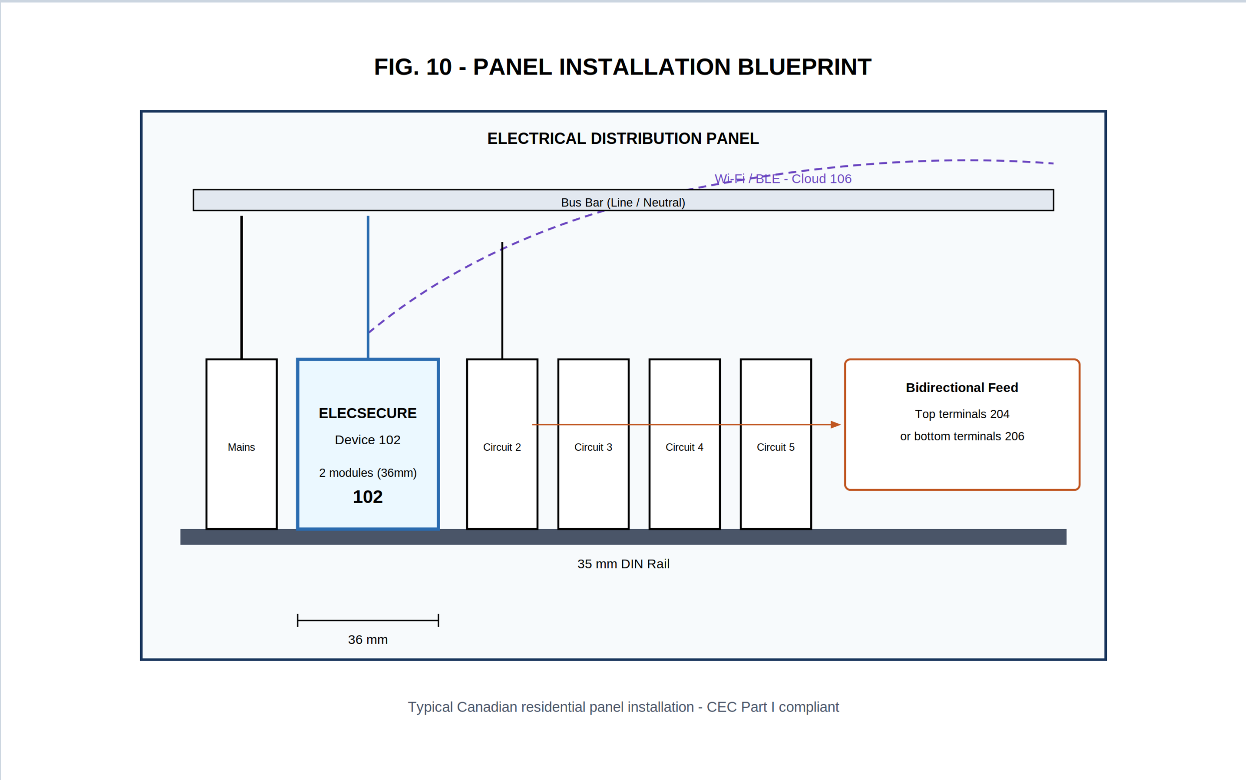


FIG. 10 - Typical Canadian residential panel installation with DIN rail mounting and bidirectional feed options.

Bill of Materials & Electrical Specifications

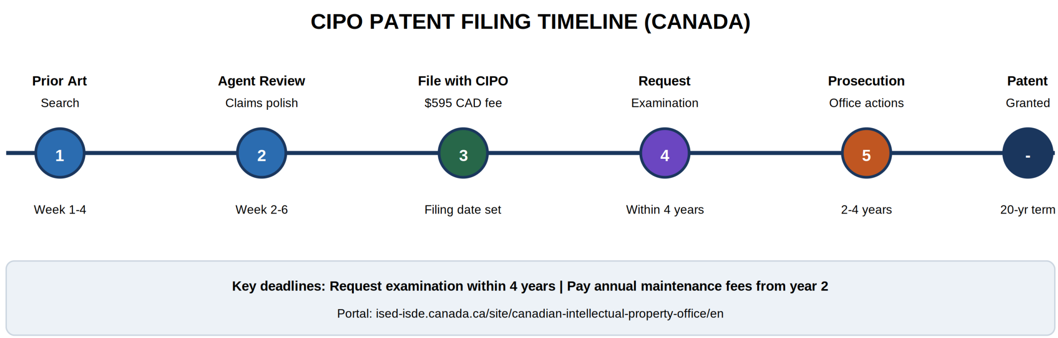
Parameter	Value / Range
Rated voltage	120/240 V AC (split-phase, Canadian residential)
Rated current	15 A / 20 A / 30 A (model variants)
Breaking capacity	6 kA or 10 kA @ rated voltage
Thermal trip	Inverse-time, IEC Type B/C curve
Magnetic trip	Instantaneous, 5–10 × In
Arc detection response	< 100 ms (target)
Wi-Fi	2.4 GHz, 802.11 b/g/n (ISED certified module)
Bluetooth	BLE 5.0 for provisioning
Operating temperature	–25 °C to +55 °C
Form factor	2-module DIN rail (36 mm width)
Ingress protection	IP20 (panel interior)

Key Components (BOM)

Ref	Component	Qty
U1	Microcontroller (ARM Cortex-M4, 120 MHz)	1
U2	Wi-Fi/BLE combo module (ISED-certified)	1
T1	Current transformer / shunt (0–30 A)	1
R1	Voltage sensing divider network	1
TH1	NTC temperature sensor	1
AS1	Arc sensor (optical / HF current)	1–2
K1	Magnetic trip solenoid	1
BM1	Bimetallic thermal element	1
AR1	Arc runner assembly (copper alloy)	1
PC1	Pre-chamber plate set (ceramic)	2–4
AC1	Arc chamber housing (vented)	1
LED1–4	Status LEDs (fault categories)	4

Ref	Component	Qty
PS1	AC-DC power supply IC (3.3 V / 5 V)	1
M1	Flash memory (4–16 MB)	1

CIPO Filing Roadmap



Fee item	Standard entity	Small entity (if eligible)
Application (filing)	$595.06 CAD	$241.24 CAD
Request for examination	$1,190.13 CAD	$482.48 CAD
Excess claims (>20)	$117.94 CAD each	$58.97 CAD each
Final fee (on allowance)	$446.03 CAD	$181.20 CAD

Applicant Declaration & Signature

I declare that the information in this patent application package is true and complete to the best of my knowledge. I understand this is a draft document and must be reviewed by a registered Canadian patent agent before filing with CIPO.

Inventor / Applicant	Azfar Mushtaq
Signature	____
Date	____
Address	____
City, Province, Postal Code	____
Patent Agent (if appointed)	____
Agent registration #	____

Filing checklist: ■ Prior art search ■ Agent review ■ Petition completed ■ Drawings finalized ■ Fees prepared ■ CIPO e-filing account ■ Examination request scheduled